General Chemistry 1, Chemistry NNNN
Fall 2027

Instructor: Md. Sharear Saon **Room:** Somewhere **Email:** sharearsaon@outlook.com **Phone:** (314)500-3617	**Class schedule:** MWF 10:00 am – 10:50 am **Class location:** Somewhere **Office hours:** Friday 3:00 pm – 4:00 pm Office hours location: Somewhere or zoom

Course Description
This is the first part of two introductory courses for essential concepts of chemistry. Theories, laws, and definitions explained in this course will be helpful to explore more specific branches of chemistry, for instance, inorganic, organic, physical, analytical, or biochemistry. Previous experience in basic algebra will be helpful to solve problems and deduce logical conclusions using multiple theories or laws. This course will mainly focus on different parameters to understand and predict the physical and chemical properties of different compounds.

Goals
1. Introducing the structural features of molecules and atoms.
2. Students will be introduced to some important chemical reactions with real-world significance.
3. Explaining the significance of different thermodynamic parameters.
4. This course will introduce different chemical bonds and their relationship with the physical and chemical properties of substances.

Objectives
1. Students will understand the structure of different molecules.
2. Students will be able to predict the stability of different molecules towards physical and chemical change.
3. Students will be able to distinguish between different reactions taking place in the real world.
4. Students will be able to determine the impact of a chemical reaction on the environment.
5. Students will understand the different steps of a laboratory experiment.
6. Students will be able to interpret and analyze scientific data.

Required Course Materials
1. *General Chemistry*, 11th Edition
   Authors: Darrell Ebbing and Steven D. Gammon
   ISBN-13: 978-1305580343
   ISBN-10: 1305580346
2. Non-programmable scientific calculator
3. iClicker 2
   You can purchase this from bookstore. You need to register your iClicker on your Canvas account by September 7th.
4. Canvas account through MySLU tools.
5. Computer with internet connection to access course materials.

Technology used in the course
- PowerPoint
  - Lecture will be presented in the class and posted in Canvas as PowerPoint format

- iClicker
    - Posing questions to create interactive learning environments
- Canvas
    - Posting course materials (announcements, grades, assignments, feedback)
    - Posting questions (anyone can post and respond to another post)
    - Interactive reading assignment (highlight, post questions, respond to other's questions)
- Google Drive
    - Additional materials will be shared via Google Drive links
    - Google Sheets can be used to conduct group activities in the class

<u>Assessments</u>

1. **iClicker questions (50 points, 5% of the total)**
    - Each lecture will contain at least one iClicker question.
    - Questions can be asked anytime in the lecture period.
    - iClicker questions will be asked to remind an old topic or practice one discussing in the class.
    - If a student cannot attend a class, they will miss the points for iClicker questions that day.
2. **Online quizzes (70 points, 7% of the total)**
    - Before starting a new chapter
    - 5 multiple choice questions from the new chapter
    - Quiz will be posted online 1 day before the new chapter starts
    - Total 14 online quizzes will be posted online, 13 for each chapter and one for surveying student's expectation. For the later one, student will be awarded full points just to complete.
    - Due dates will be reminded over email and Canvas notification
3. **In-class activities (90 points, 9% of the total)**
    - There will be 9 in-class activities (10 points for each).
    - In-class activities can be formatted as individual or group activities.
    - To earn the full point students will require just to participate.
    - Questions or problems will be posed to practice the topic introduced in the class.
4. **Reading assignment (50 points, 5% of the total)**
    - 5 Reading assignments will be posted on Canvas (10 points for each).
    - Due dates will be announced in the class and in the Canvas homepage as well.
    - Students will require to post a certain number (will be determined later) of comments/questions/answers for other questions to get the full credit.
5. **In-class quizzes (200 points, 20% of the total)**
    - There will be 10 short quizzes (20 points for each and one after each chapter).
    - Each quiz will contain 10 questions (2 points for each).
    - Quiz will take place in the class right after finishing one chapter.
    - There will be 10 questions (Format of questions: multiple choice questions, fill in the blanks, short answers) and 15 minutes will be provided.
6. **Homework assignment (100 points, 10% of the total)**
    - Throughout the semester there will be 5 homework assignments (20 points for each).
    - Assignment will contain practice problems to help students revisit the important concepts of the preceding chapters.

- Every third Friday at 5:00 pm one assignment will be posted on canvas.
- Students need to submit their work in canvas by the following Monday no later than 5:00 pm.

7. **Exams (440 points, 44% of the total)**
    - There will be five exams throughout the semester.
    - 85 points for the first four exams and 100 points for the final exam.
    - Exam will contain multiple choice questions, short and broad questions.
    - Except for the multiple-choice questions, students' responses will be graded partially if the final answer is not as expected.
    - The first four exam will be one hour long, and the final exam will be one and half hour long.
    - Students should leave their smart watch and smart phone in their backpack which can be stored at the front of the class.
    - With proper medical proof students can take the exam online, however, the entire time they should leave their camera turned on.
    - The materials for the final exam will be covered from the entire course.
    - To encourage students to continue their improvement and learning, if a student's final grade is higher than the average of the other four exams, then the lowest grade among the other four will be replaced with the average between the final grade and the original grade of that exam.

Grades

In the final result, letter grade will be assigned using the following schema.

Letter grade	Point range	%	Letter grade	Point range	%
A	900-1000	90.00-100.00%	C+	650-699	65.00-69.99%
A-	850-899	85.00-89.99%	C	600-649	60.00- 64.99%
B+	800-849	80.00-84.99%	C-	550-599	55.00- 59.99%
B+	750-799	75.00- 79.99%	D	450-549	45.00 - 54.99%
B-	700-749	70.00- 74.99%	F	<450	<45.00%

Academic Integrity

Academic integrity is honest, truthful and responsible conduct in all academic endeavors. The mission of Saint Louis University is "the pursuit of truth for the greater glory of God and for the service of humanity." Accordingly, all acts of falsehood demean and compromise the corporate endeavors of teaching, research, health care, and community service through which SLU embodies its mission. The University strives to prepare students for lives of personal and professional integrity and therefore regards all breaches of academic integrity as matters of serious concern.

The governing University-level Academic Integrity Policy was adopted in Spring 2015, and can be accessed on the Provost's Office website (http://www.slu.edu/provost/policies.php).

Additionally, each SLU college, school and center has adopted its own academic integrity policies, available on their respective websites. All SLU students are expected to know and abide by these policies, which detail definitions of violations, processes for reporting violations, sanctions, and appeals. Please direct questions about any facet of academic integrity to your faculty, the chair of the department of your academic program, or the dean/director of the college, school or center in which your program is housed.

Specific College of Arts and Sciences Academic Honesty Policies and Procedures may be found here (http://www.slu.edu/arts-and-sciences/student-resources/academic-honesty.php).

Title IX

Saint Louis University and its faculty are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of sexual misconduct (e.g. sexual assault, sexual harassment, stalking, domestic or dating violence), we encourage you to report this to the University. If you speak with a faculty member about an incident of misconduct, that faculty member must notify SLU's Title IX coordinator, Anna R. Kratky (DuBourg Hall, room 36; anna.kratky@slu.edu; 314-977-3886) and share the basic facts of your experience with her. The Title IX coordinator will then be available to assist you in understanding all of your options and in connecting you with all possible resources on and off campus.

If you wish to speak with a confidential source, you may contact the counselors at the University Counseling Center at 314-977-TALK. To view SLU's sexual misconduct policy and for resources, please visit the following web addresses:
http://www.slu.edu/about/safety/pdfs/sexual-misconduct-policy-version7.0.pdf.

Student Success Center

In recognition that people learn in a variety of ways and that learning is influenced by multiple factors (e.g., prior experience, study skills, learning disability), resources to support student success are available on campus. The Student Success Center assists students with academic and career related services, is located in the Busch Student Center (Suite, 331) and the School of Nursing (Suite, 114). Students who think they might benefit from these resources can find out more about:

·Course-level support (e.g., faculty member, departmental resources, etc.) by asking your course instructor.
·University-level support (e.g., tutoring services, university writing services, disability services, academic coaching, career services, and/or facets of curriculum planning) by visiting the Student Success Center.

Disability Services Academic Accommodations

Students with a documented disability who wish to request academic accommodations are encouraged to contact Disability Services to discuss accommodation requests and eligibility requirements.

Please contact Disability Services, located within the Student Success Center, at disability_services@slu.edu or 314.977.3484 to schedule an appointment. Confidentiality will be observed in all inquiries.
Once approved, information about the student's eligibility for academic accommodations will be shared with course instructors via email from Disability Services and viewed within Banner via the instructor's course roster.

Course Schedule

Week	Date	Day	Lecture content
Week 1	August 30, 2027	Monday	Course Introduction, Chapter 1: Chemistry and Measurement
	September 1, 2027	Wednesday**	Chapter 1 (Cont), Chapter 2: Atoms, Molecules, and Ions
	September 3, 2027	Friday**	Chapter 2 (Cont), Chapter 3: Calculations with Chemical Formulas
Week 2	September 6, 2027	Monday	Labor day (No class)
	September 8, 2027	Wednesday	Chapter 3 (Cont)
	September 10, 2027	Friday	Chapter 3 (Cont)
Week 3	September 13, 2027	Monday**	Chapter 3 (Cont), Chapter 4: Chemical Reactions
	September 15, 2027	Wednesday	Chapter 4 (Cont)
	September 17, 2027	Friday*	Chapter 4 (Cont)
Week 4	September 20, 2027	Monday**	Chapter 4 (Cont), Chapter 5: The Gaseous State
	September 22, 2027	Wednesday	Chapter 5 (Cont)
	September 24, 2027	Friday	Chapter 5 (Cont)
Week 5	September 27, 2027	Monday	**Exam 1**
	September 29, 2027	Wednesday	Questions review from Exam 1, Chapter 5 (Cont)
	October 1, 2027	Friday**	Chapter 5 (Cont), Chapter 6: Thermochemistry
Week 6	October 4, 2027	Monday	Chapter 6 (Cont)
	October 6, 2027	Wednesday	Chapter 6 (Cont)
	October 8, 2027	Friday* **	Chapter 6 (Cont), Chapter 7: Quantum Theory of the Atom
Week 7	October 11, 2027	Monday	Chapter 7 (Cont)
	October 13, 2027	Wednesday	Chapter 7 (Cont)
	October 15, 2027	Friday**	Chapter 7 (Cont), Chapter 8 :Electron Configurations and Periodicity
Week 8	October 18, 2027	Monday	Chapter 8 (Cont)
	October 20, 2027	Wednesday	Chapter 8 (Cont)
	October 22, 2027	Friday	Fall break (No class)
Week 9	October 25, 2027	Monday	**Exam 2**
	October 27, 2027	Wednesday	Questions review from Exam 2, Chapter 8 (Cont)

Week	Date	Day	Lecture content
	October 29, 2027	Friday* **	Chapter 8 (Cont), Chapter 9: Ionic and Covalent Bonding
Week 10	November 1, 2027	Monday	Chapter 9 (Cont)
	November 3, 2027	Wednesday	Chapter 9 (Cont)
	November 5, 2027	Friday**	Chapter 9 (Cont), Chapter 10: Molecular Geometry and Chemical Bonding Theory
Week 11	November 8, 2027	Monday	Chapter 10 (Cont)
	November 10, 2027	Wednesday	Chapter 10 (Cont)
	November 12, 2027	Friday	Chapter 10 (Cont)
Week 12	November 15, 2027	Monday**	Chapter 10 (Cont), Chapter 11 (States of Matter; Liquids and Solids)
	November 17, 2027	Wednesday	Chapter 11 (Cont)
	November 19, 2027	Friday*	Chapter 11 (Cont)
Week 13	November 22, 2027	Monday	**Exam 3**
	November 24, 2027	Wednesday**	Chapter 11 (Cont), Chapter 12: Solutions
	November 26, 2027	Friday	Chapter 12 (Cont)
Week 14	November 29, 2027	Monday	Chapter 12 (Cont)
	December 1, 2027	Wednesday	Chapter 12 (Cont)
	December 3, 2027	Friday* **	Chapter 12 (Cont), Chapter 13: Rates of Reaction
Week 15	December 6, 2027	Monday	Chapter 13 (Cont)
	December 8, 2027	Wednesday	Chapter 13 (Cont)
	December 10, 2027	Friday	Chapter 13 (Cont)
Week 16	December 13, 2027	Monday	**Exam 4**
	December 15, 2027	Wednesday	Review
	December 17, 2027	Friday	No Class
	December 20, 2027	Monday	**Final Exam**

* Assignment will be posted in Canvas
** Quiz (covered the chapter that was just finished) day